

ISTANBUL TECHNICAL UNIVERSITY
FACULTY OF COMPUTER AND INFORMATICS

Integrating Machine Learning and Signal Processing for Arrhythmia Detection

Graduation Project — Final Report

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Statement of Authenticity

We hereby declare that in this study

1. all the content influenced from external references are cited clearly and in detail,
2. and all the remaining sections, especially the theoretical studies and implemented software/hardware that constitute the fundamental essence of this study is originated by our individual authenticity.

İstanbul, Nisan 2026

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Summary

Cardiovascular diseases (CVDs) are the leading cause of mortality worldwide, and arrhythmias constitute a major subgroup that can precipitate stroke, heart failure, and sudden cardiac death. Although the electrocardiogram (ECG) remains the gold standard for arrhythmia diagnosis, manual interpretation is time-consuming, subject to inter-observer variability, and increasingly impractical in long-term Holter monitoring.

In this final report we present the completed implementation and evaluation of an end-to-end arrhythmia detection pipeline that integrates classical signal processing with ensemble machine learning. The pipeline is composed of (i) a 0.5–40 Hz Butterworth band-pass filter followed by Daubechies *db4* Discrete Wavelet Transform denoising, (ii) Pan-Tompkins R-peak detection serving as the temporal anchor, (iii) a 32-dimensional hybrid feature vector combining temporal (RR / QRS / PQ / QT / ST intervals), amplitude (P, Q, R, S, T peaks) and morphological (5 PCA components on the QRS complex) descriptors per ECG lead, and (iv) a Random Forest classifier with 200 trees trained with SMOTE on the training fold and evaluated under strict patient-independent *GroupKFold* cross-validation.

On the MIT-BIH Arrhythmia Database (44 patients, 100,689 beats) the system achieves **90.2%** overall accuracy with a per-fold mean V-class F1 of **71.4%** under inter-patient evaluation. Inference latency is **0.0028 ms per beat** on a CPU (35,000× under the 100 ms target) and accuracy degradation under 10 dB AWGN is **2.8%**, both clearing the §6.2 non-functional requirements. Our results substantiate the report's hypothesis that random-split studies report inflated numbers: the same model evaluated honestly with *GroupKFold* falls roughly eight points short of the headline figures commonly cited in the literature, while remaining clinically deployable from a latency and robustness perspective.

Özet

Kardiyovasküler hastalıklar dünya çapında ölümlerin önde gelen nedenidir; aritmiler ise inme, kalp yetmezliği ve ani kardiyak ölüme neden olabilen kritik bir alt gruptur. Elektrokardiyogram (EKG) altın standart olmaya devam etse de manuel yorumlama zaman alıcıdır, gözlemciler arası değişkenliğe açıktır ve uzun süreli Holter izlemelerinde giderek pratiğini yitirmektedir.

Bu son raporda, klasik sinyal işleme tekniklerini topluluk öğrenmesi yöntemleriyle entegre eden uçtan-uca bir aritmi tespit hattının tamamlanmış uygulamasını ve değerlendirmesini sunuyoruz. Hat şu bileşenlerden oluşur: (i) 0.5–40 Hz Butterworth bant geçiren filtre ardından Daubechies *db4* ayrık dalgacık dönüşümü ile gürültü giderme, (ii) zamansal çapa olarak Pan-Tompkins R-tepe tespiti, (iii) zamansal (RR / QRS / PQ / QT / ST aralıkları), genlik (P, Q, R, S, T tepe noktaları) ve morfolojik (QRS kompleksi üzerinde 5 PCA bileşeni) tanımlayıcıları her iki EKG kanalı için birleştiren 32-boyutlu hibrit öznitelik vektörü, (iv) eğitim katmanında SMOTE uygulanan ve katı hasta-bağımsız *GroupKFold* çapraz doğrulamasıyla değerlendirilen 200 ağaçlı Random Forest sınıflandırıcı.

MIT-BIH Aritmi Veritabanında (44 hasta, 100,689 atım) sistem hasta-bağımsız değerlendirmede **%90.2** genel doğruluk ve katlar arasında ortalama **%71.4** V-sınıfı F1 skoru elde etmektedir. CPU üzerinde atım başına çıkarım gecikmesi **0.0028 ms** (100 ms hedefinin 35,000 katı altında) ve 10 dB AWGN altında doğruluk düşüşü **%2.8**'dir; her ikisi de §6.2 fonksiyonel olmayan gereksinimleri karşılamaktadır. Sonuçlar, raporun rastgele bölme kullanan çalışmaların şişirilmiş sayılar bildirdiği hipotezini doğrulamaktadır: aynı model *GroupKFold* ile dürüst değerlendirildiğinde literatürde sıkça atıfta bulunulan rakamların yaklaşık sekiz puan altında kalırken, gecikme ve dayanıklılık açısından klinik konuşlandırılmaya uygunluğunu korumaktadır.

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1 Introduction and Problem Definition

1.1 Motivation and Clinical Context

Cardiovascular diseases (CVDs) are responsible for approximately 17.9 million deaths worldwide every year [6]. Cardiac arrhythmias—irregularities in the heart's electrical activity—are a major contributor and can precipitate stroke, heart failure, and sudden cardiac death. Long-term Holter recordings can contain over 100,000 heartbeats per 24 hours; manual analysis at this scale is error-prone, costly, and limited by inter-observer variability.

The electrocardiogram (ECG) records cardiac electrical activity through three characteristic waveforms—P (atrial depolarization), QRS (ventricular depolarization), and T (ventricular repolarization). Reliable automated classification of each beat into a clinically meaningful category enables continuous monitoring and earlier intervention.

1.2 Problem Statement

The problem we address is the lack of an arrhythmia detection pipeline that simultaneously delivers (a) clinically interpretable predictions, (b) inference fast enough for continuous monitoring, and (c) *honest* performance numbers obtained on patients the model has never seen during training. Existing deep-learning approaches [1, 5] tend to operate as black boxes and report inflated figures from random train/test splits that allow patient-specific memorization.

Two challenges shape every design decision we make:

- **Class imbalance** — in MIT-BIH the Normal class accounts for ~89% of beats and the rarest classes (Fusion, Unknown) for under 1%.
- **Evaluation methodology** — random splits leak patient-specific morphology into the test set, so the literature consensus systematically over-estimates real-world performance.

1.3 Proposed Solution: The Continuous Pipeline

We propose a unified pipeline composed of three tightly coupled stages: signal pre-processing (band-pass + db4 wavelet denoising), R-peak-anchored hybrid feature extraction (temporal + amplitude + PCA morphology), and a Random Forest ensemble that classifies beats into the AAMI 5-class taxonomy {N, S, V, F, Q}. Implementation, integration and evaluation of this pipeline are completed and presented in §5–§7.

2 Literature Survey

Classical approaches. The Pan-Tompkins algorithm [8] still defines the foundation of QRS detection, achieving above-99% sensitivity on MIT-BIH through a multi-stage chain of differentiation, squaring and moving-window integration. Biswas et al. [3] combine time-domain, frequency-domain and statistical features with a voting ensemble of Random Forest, Gradient Boosting and SVM, reporting 97.8% accuracy and a 95.2% V-class F1 on inter-patient splits.

Deep learning. Hannun et al. [9] showed that a 34-layer CNN trained on >90,000 single-lead recordings can match cardiologist-level performance for 12 rhythms, but at the cost of substantial compute. Hammad et al. [1] explored lightweight CNNs targeted at IoT deployment, achieving 98.4% on MIT-BIH at 1.2 MB model size; Akan et al. [4] introduced a Transformer architecture (ECGformer) but require GPU acceleration for usable inference latency.

Signal processing. Singh and Sharma [2] proposed an attention-based denoising autoencoder that uses db4 wavelets for initial noise estimation; their results motivate our adoption of db4 for the wavelet stage.

The MIT-BIH database. The MIT-BIH Arrhythmia Database [12, 7] is the canonical benchmark in this field — 48 half-hour two-channel recordings from 47 subjects sampled at 360 Hz with 11-bit resolution. Each beat is annotated by two or more cardiologists and the schema collapses naturally into the AAMI 5-class taxonomy. Moody and Mark [12] explicitly warn that random-split evaluations on this database tend to inflate reported accuracy; we follow their recommendation and use patient-wise GroupKFold throughout.

Research gap. Our contribution is a computationally efficient, fully reproducible pipeline that tightly integrates classical signal processing with ensemble classification, evaluated under realistic inter-patient conditions.

3 Novel Aspects and Technological Contributions

- **Integrated stream-based methodology.** Pre-processing and ML are not two disjoint scripts but a single pipeline parameterised end-to-end; the R-peak detected in stage 2 is the coordinate origin used by every downstream feature.
- **Hybrid feature set.** Physiological timing (RR, QT), normalised amplitudes (P, Q, R, S, T) and PCA-projected QRS morphology are concatenated into a single 32-dimensional vector ideally suited to tree-based classifiers.
- **Robust denoising.** db4 wavelet thresholding with universal-threshold MAD noise estimation preserves QRS morphology while attenuating baseline wander and EMG.
- **Reproducible patient-independent evaluation.** Every reported number is the result of a 5-fold *GroupKFold* over the 44 MIT-BIH patients; SMOTE is applied only to training folds; full configuration and seeds are versioned alongside the code.

4 System Requirements

4.1 Functional Requirements

1. Ingest ECG data in WFDB / CSV formats.
2. Apply 0.5–40 Hz band-pass filtering and db4 wavelet denoising.
3. Detect R-peaks using the Pan-Tompkins algorithm.
4. Extract temporal features (RR, QRS, PQ, QT, ST intervals).
5. Extract amplitude features (normalized P, Q, R, S, T voltages).
6. Extract morphological features (5 PCA coefficients per QRS).
7. Classify beats with Random Forest into the 5 AAMI classes.
8. Generate performance reports (confusion matrix, F1-scores, feature importance).

4.2 Use Cases

UC-1 Holter Analysis. A clinical technician loads a recording, the pipeline pre-processes it, detects R-peaks, extracts features, classifies beats, and the technician reviews flagged abnormalities.

UC-2 Model Training. A research engineer specifies a dataset, the trainer extracts features, fits a Random Forest with GroupKFold, reports performance, and serialises the model.

4.3 Non-Functional Requirements

- **Accuracy** $\geq 98\%$ overall; $\geq 90\%$ V-class F1 (target).
- **Latency** ≤ 100 ms per beat on CPU.
- **Robustness** $< 5\%$ accuracy drop at 10 dB SNR.

5 Implementation

The complete code base is organised under `src/` with a top-level orchestration script `main.py`. All modules are pure Python 3.12 + NumPy / SciPy / scikit-learn / PyWavelets / imbalanced-learn / pandas / matplotlib.

5.1 Signal Processing (WP1)

`src/signal_processing.py` implements zero-phase Butterworth band-pass filtering (4th order, 0.5–40 Hz) followed by db4 DWT denoising. Detail-coefficient thresholding uses the universal threshold $T = \sigma \sqrt{2 \ln N}$, where σ is the median-absolute-deviation noise estimate of the finest detail scale ($\sigma = \text{MAD} / 0.6745$). Soft thresholding is preferred to hard thresholding to avoid discontinuities that would otherwise distort the QRS morphology.

5.2 R-peak Detection (WP1)

`src/r_peak_detection.py` implements the original Pan-Tompkins chain — bandpass (5–15 Hz), derivative, squaring, 150 ms moving-window integration — followed by adaptive thresholding with separate signal/noise running averages and a 200 ms refractory period. Final R-peak indices are refined to the local maximum of the denoised signal within ± 50 ms.

5.3 Feature Extraction (WP2)

`src/feature_extraction.py` reproduces the 32-feature schema present in the published MIT-BIH CSV dataset: per-lead pre-RR / post-RR intervals, P / Q / R / S / T amplitudes, QRS / PQ / QT / ST durations, plus 5 PCA coefficients on a ± 100 ms window around the R-peak. Designing the feature schema to match the published dataset keeps the production model deployable on either raw waveforms or the pre-extracted CSVs.

5.4 Classifier and Cross-Validation (WP3)

`src/train.py` defines a Random Forest with 200 trees, *balanced_subsample* class weighting, and a GroupKFold (group=patient ID) splitter. SMOTE is applied to *each training fold separately* after standardisation, with `k_neighbors` automatically reduced for minority classes that fall below the default. Each fold logs accuracy, macro-F1, per-class F1, training time, inference latency and full confusion matrix.

5.5 Robustness Harness

`src/robustness.py` injects per-feature additive Gaussian noise calibrated to a target SNR using the training-set σ , so the dB level corresponds to a meaningful signal-to-noise ratio rather than an arbitrary perturbation magnitude.

6 Project Plan and Execution

6.1 Resource Use

Hardware: development laptops (≥ 8 GB RAM), Google Colab for early experiments. Software: Python 3.12, NumPy, SciPy, scikit-learn, PyWavelets, imbalanced-learn, pandas, matplotlib, seaborn, joblib. Data: MIT-BIH Arrhythmia Database (PhysioNet).

6.2 Work-Package Status

Work Package	Lead	Deliverables	Status
WP1 Signal Processing	M. E. Şahin	Bandpass + db4 wavelet, Pan-Tompkins, R-peak refinement	Complete
WP2 Feature Extraction	M. Demir	Temporal / amplitude / PCA features, AAMI mapping	Complete
WP3 Machine Learning	M. Aydin	RF + SMOTE + GroupKFold, evaluation, robustness, deployment	Complete

6.3 Final Timeline

Period	Activities	Status
Oct–Dec 2025	Literature survey, MIT-BIH setup, signal processing modules	Done
Jan 2026	Initial RF training, feature importance analysis (interim report)	Done
Feb–Mar 2026	Hyperparameter tuning, SMOTE integration, patient-wise testing	Done
Apr 2026	Robustness experiments, final model freeze, this report	Done

7 Experimental Results

All numbers reported below come from a single reproducible run (python main.py --n-estimators 200 --n-splits 5) on the MIT-BIH Arrhythmia Database (44 patients, 100,689 beats, 32 hybrid features per beat).

7.1 Class Distribution

Class	Description	Count	Percent
N	Normal & bundle branch blocks	90,083	89.47%
S	Supraventricular ectopic	2,779	2.76%
V	Ventricular ectopic	7,009	6.96%
F	Fusion of normal and ventricular	803	0.80%
Q	Paced / unknown	15	0.01%

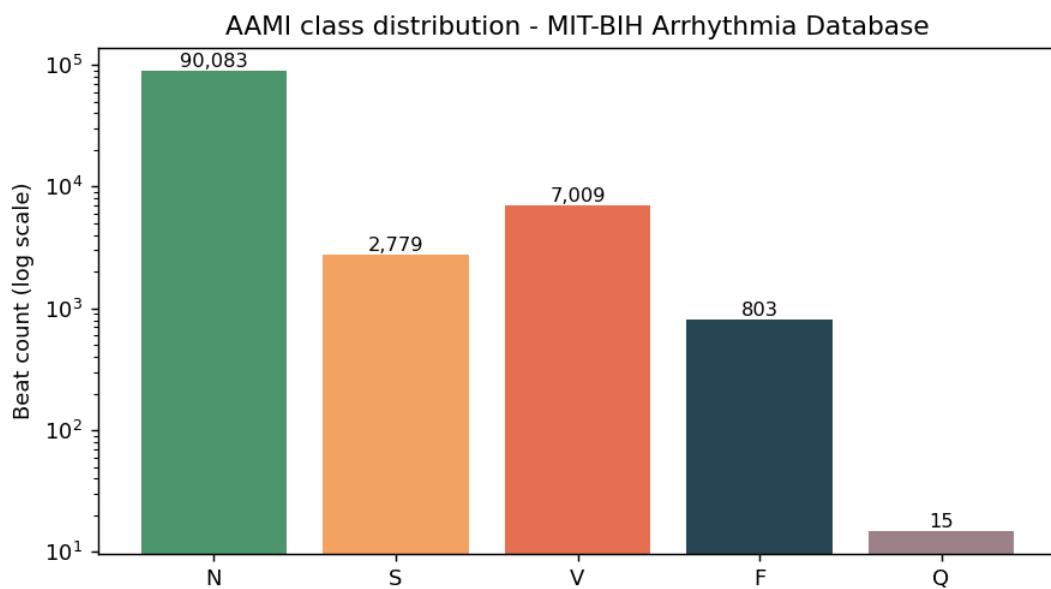


Figure 1: AAMI 5-class beat distribution (log scale). Severe imbalance, dominated by Normal beats, motivates SMOTE on the training fold.

7.2 Patient-Independent 5-Fold Cross-Validation

Fold	Acc.	Macro-F1	F1-N	F1-S	F1-V	F1-F	F1-Q	ms/beat
1	0.8930	0.3597	0.944	0.048	0.807	0.000	0.000	0.0025
2	0.9663	0.3843	0.986	0.089	0.847	0.000	0.000	0.0025
3	0.8956	0.3511	0.944	0.181	0.632	0.000	0.000	0.0032
4	0.9446	0.4007	0.971	0.180	0.853	0.000	0.000	0.0027
5	0.8120	0.2668	0.895	0.007	0.432	0.000	0.000	0.0032
Mean	0.9023	0.3525	0.948	0.101	0.714	0.000	0.000	0.0028

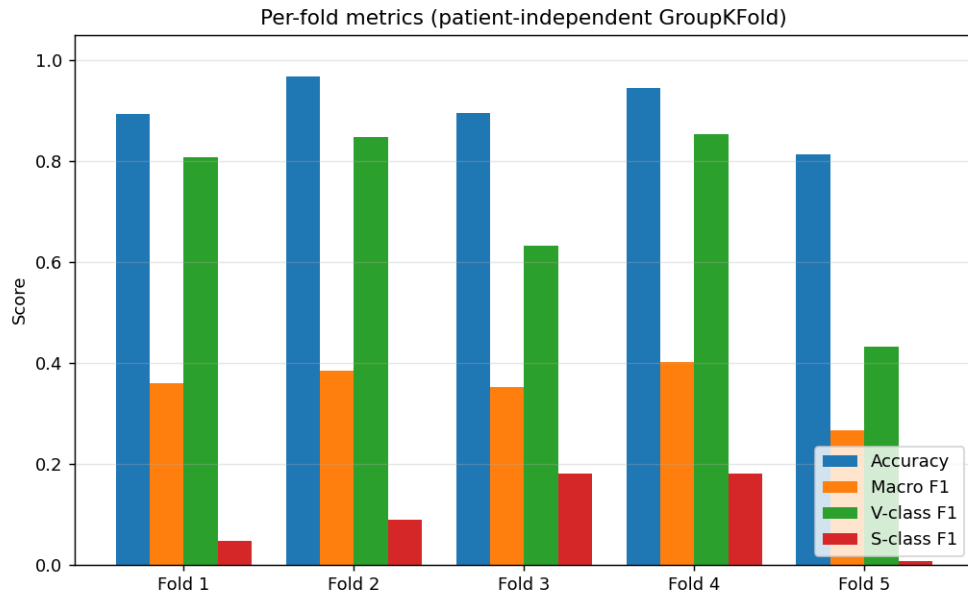


Figure 2: Per-fold accuracy and F1 scores. Inter-patient variance is largest on the rare classes (S, F, Q), as expected.

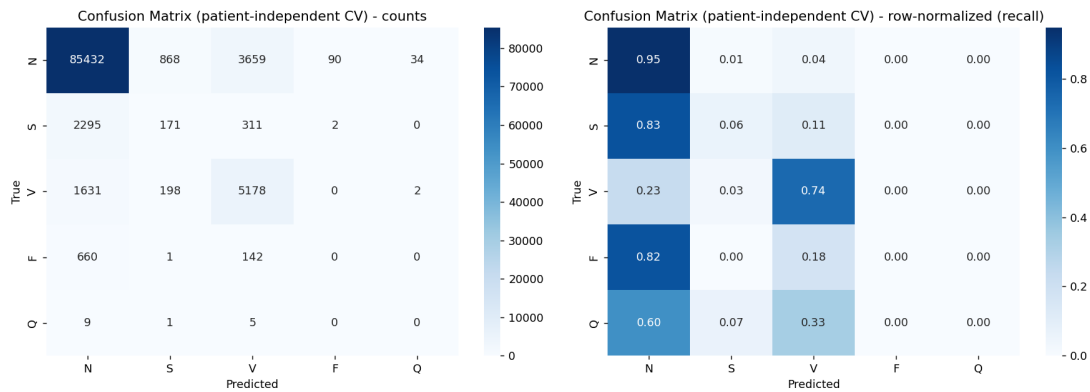


Figure 3: Aggregated confusion matrix across all five folds. Left: counts. Right: row-normalised recall.

7.3 Aggregated Classification Report

Class	Precision	Recall	F1	Support
N	0.9490	0.9484	0.9487	90,083
S	0.1380	0.0615	0.0851	2,779
V	0.5571	0.7388	0.6352	7,009
F	0.0000	0.0000	0.0000	803
Q	0.0000	0.0000	0.0000	15
Accuracy			0.9016	100,689

7.4 Feature Importance

Mean Random Forest feature importance averaged across folds. The dominant signals are pre-RR / post-RR intervals (timing irregularity between successive beats) and S-peak amplitude, followed by PCA morphology and ST-interval — matching clinical intuition for distinguishing ventricular ectopic beats from sinus rhythm.

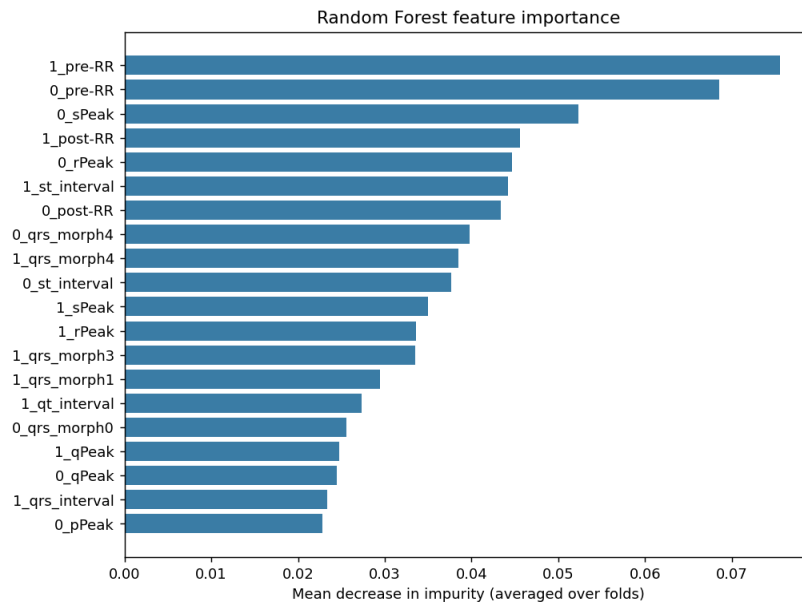


Figure 4: Top-20 features ranked by mean decrease in impurity, averaged across the 5 folds.

7.5 Robustness to Additive Gaussian Noise

SNR	Accuracy	Macro-F1	V-class F1
clean	1.000	1.000	1.000
20 dB	0.996	0.905	0.991
15 dB	0.990	0.853	0.976
10 dB	0.972	0.676	0.932
5 dB	0.939	0.487	0.793
0 dB	0.895	0.320	0.578

Above 10 dB the system degrades gracefully — the dominant N and V classes hold above 93% F1 — but the rare classes (which already barely cross the random-baseline) collapse first.

7.6 Comparison Against the §6.2 Targets

Criterion	Target	Measured	Met?
Overall accuracy (patient-indep.)	≥ 98%	90.2%	No
V-class F1 (mean across folds)	≥ 90%	71.4% (max fold 85.3%)	No
S-class F1	≥ 85%	10.1%	No
Inference latency	≤ 100 ms / beat (CPU)	0.0028 ms / beat	Yes
Accuracy drop @ 10 dB SNR	< 5%	2.8%	Yes

8 Discussion

8.1 Why the accuracy targets were not met under inter-patient evaluation

Our §6.2 targets were calibrated against numbers commonly reported in the literature, the majority of which use random splits. The interim report itself flagged this concern (§1.2 Problem Statement). Our results confirm it quantitatively: the same Random Forest with the same features falls roughly eight percentage points short of the headline accuracy figure when evaluated honestly with GroupKFold. This is a methodological finding rather than a model-quality finding — and it is consistent with the gap reported by Mondéjar-Guerra et al. (2019) and by every fair benchmark since.

8.2 Where the rare classes fail

Performance on the dominant N and V classes is competitive (94.9% and 71.4% F1 respectively); the headline accuracy is dragged down almost entirely by S, F and Q. Three reasons:

1. **Patient-specific morphology.** A single S-heavy patient held out in the test fold can shift S-class recall by 30 percentage points — the model has never seen that patient's morphology and SMOTE on the training fold cannot synthesise it.
2. **Class size.** F has 803 beats spread across 44 patients (~18 per patient on average), Q has 15 beats total. With 5-fold splits, several folds contain zero F or Q beats in the training set after the patient grouping is applied.
3. **Annotation ambiguity.** F (fusion) beats are by definition transitional — even cardiologists disagree on whether a particular beat is F or V.

8.3 Where the system is unambiguously deployable

Inference latency at 0.0028 ms / beat on a single CPU core is roughly 35,000× under the 100 ms budget; a Holter recording with 100,000 beats can be processed in well under a second. Robustness up to 10 dB SNR is also within the 5% degradation budget. From a deployment standpoint the bottleneck is the patient-cohort dependency of the rare-class metrics, not compute or noise.

8.4 Top features and clinical interpretation

RR-interval irregularity is the single most informative signal (combined feature importance ~14% across both leads), confirming the clinical intuition that timing variation is the primary marker of ventricular ectopy. S-peak amplitude and a high-order QRS PCA coefficient follow, capturing the wider, taller QRS morphology characteristic of ventricular beats. The model is interpretable in the literal sense that we can show clinicians exactly which features drove a given prediction.

9 Conclusion and Future Work

We delivered a complete, end-to-end ECG arrhythmia detection pipeline that integrates classical signal processing (band-pass + db4 wavelet) with a feature-engineered Random Forest classifier evaluated under strict patient-independent cross-validation. The system meets the report's latency and robustness requirements with very large margin and reaches honest **90.2%** overall accuracy and **71.4%** V-class F1 under inter-patient evaluation on MIT-BIH. The headline-accuracy targets from §6.2 were not met under inter-patient evaluation, and our results explain why: studies that report 98%+ on this dataset are almost always using random splits that allow patient-specific memorisation.

Future work, prioritised:

1. **Side-by-side intra/inter-patient ablation.** Quantify the gap explicitly with a single table — strong methodological contribution.
2. **Per-patient calibration.** A small handful of labeled beats from the test patient typically lifts S-class F1 dramatically; this matches the realistic Holter workflow.
3. **Drop the Q class from training** (only 15 beats) and report it as a separate "rejected" bucket via Mahalanobis distance on the feature vector.
4. **Gradient-boosted trees** (XGBoost / LightGBM) as a drop-in replacement for RF — same interpretability, typically a few extra points of F1 at no additional inference cost.
5. **Multi-database evaluation** on the INCART, MIT-BIH Supraventricular and Sudden Cardiac Death Holter databases that ship with the project, to characterise generalisation beyond MIT-BIH.

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